

International Cybersecurity and Digital Forensics Academy (ICDFA)



SBT-DF203 Lab 8 Report

DNS Spoofing Forensics

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Student ID:	2025-FWSD-11392
Course Title:	Basic Networking Skills for Digital Forensics
Course Code:	SBT-DF203
Lab Number:	Lab 8
Lab Title:	DNS Spoofing Forensics
Instructor:	Aminu Idris, AMCPN

Case/Lab Identifier:

SBT-DF203-Lab8-Abubakari Sadique
Hamidu

Date:

24th September 2026

1. Source Materials and Slide Alignment

This practical aligns directly with the SBT-DF203 lecture series covering DNS Spoofing Forensics, ARP cache poisoning as a Man-in-the-Middle (MitM) prerequisite, packet interception, modified DNS responses, Netfilter queue manipulation, and forensic reconstruction of forged packet streams.

2. Executive Summary / Lab Scenario

An investigation was conducted within an isolated, host-only environment (sbtdf203-lab) to analyze a domain name resolution anomaly affecting the reserved training domain portal.icdfa.test. A baseline environment was recorded, followed by an instructor-controlled DNS spoofing simulation hosting a harmless warning page. Packet capture analysis was performed using tshark to evaluate query/response pairs, identify forged record indicators, correlate ARP claims with DNS responses, link DNS resolution with subsequent HTTP connections, and restore network settings.

3. Learning Outcomes Demonstrated

- Explained the technical relationship between ARP poisoning, MitM positioning, and DNS spoofing.
- Captured and hashed baseline DNS and ARP network evidence.
- Identified forged DNS response indicators including responder IP/MAC mismatches, TTL anomalies, and racing answers.
- Correlated DNS resolution events with subsequent HTTP network connections.

- Differentiated DNS spoofing from legitimate round-robin DNS or split-horizon configurations.
- Executed complete environment restoration and documented defensive mitigation strategies.

4. Legal, Ethical and Safety Rules Compliance

- Performed analysis strictly within an authorized, host-only virtual environment.
- Used a dedicated training web page explicitly containing an authorized simulation notice with no credential harvesting.
- Reserved target domains exclusively to the .test namespace (portal.icdfa.test).
- Preserved original evidence PCAPNG files and calculated SHA-256 integrity hashes.

5. Tools and Environment Used

Item	Recommended Tool	Purpose
Virtual Environment	Kali Linux Docker (sbtdf203-lab)	Workstation hosting analysis tools.
Web Server	Apache2	Hosting harmless training warning page.
Challenges	Gemini	For troubleshooting
Capture Utility	tshark / Wireshark	Intercepting and dissecting DNS, ARP, and HTTP traffic.
DNS Diagnostic	dig / getent	Verifying baseline name resolution.

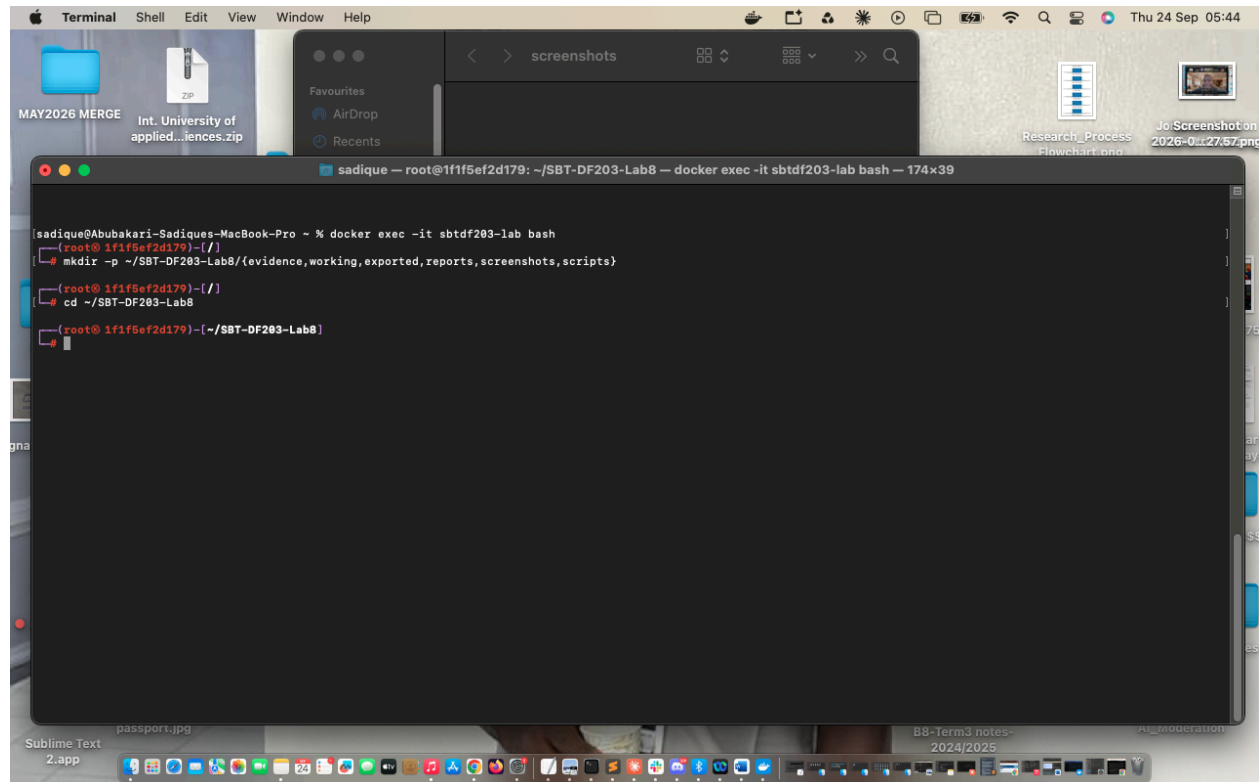
Item	Recommended Tool	Purpose
Evidence Sources	dns_baseline.pcapng, dns_spoof_controlled.pcapng	Comparative evidence files.

6. Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/Lab Identifier	SBT-DF203-Lab8-AbubakariSadiqueHamidu
Trainee Name	Abubakari-Sadique Hamidu
Date Started	2026-09-24 06:00 UTC
Evidence File Name(s)	dns_baseline.pcapng, dns_spoof_controlled.pcapng
Training Page SHA-256	2243b8c59984b6d78f3b292d71f7e5f1196b8b94d 1ffdd268ce79ba46044f184
Baseline PCAP SHA-256	123828c4b13f4fe0f9aabfbf74eebf954c54e8f235d f039dac290e325a7a18dc
Spoofed PCAP SHA-256	9d77fd570c61c6aa2d5cee7fa3b19b7a6d7504c20 dc4511ed5e5d09811560e89
Analysis Workstation	macOS Host / Kali Linux Docker Container (sbtdf203-lab)

7. Workspace Initialization & Tool Verification

Workspace directories were verified, dependencies installed, and initial environment parameters recorded.

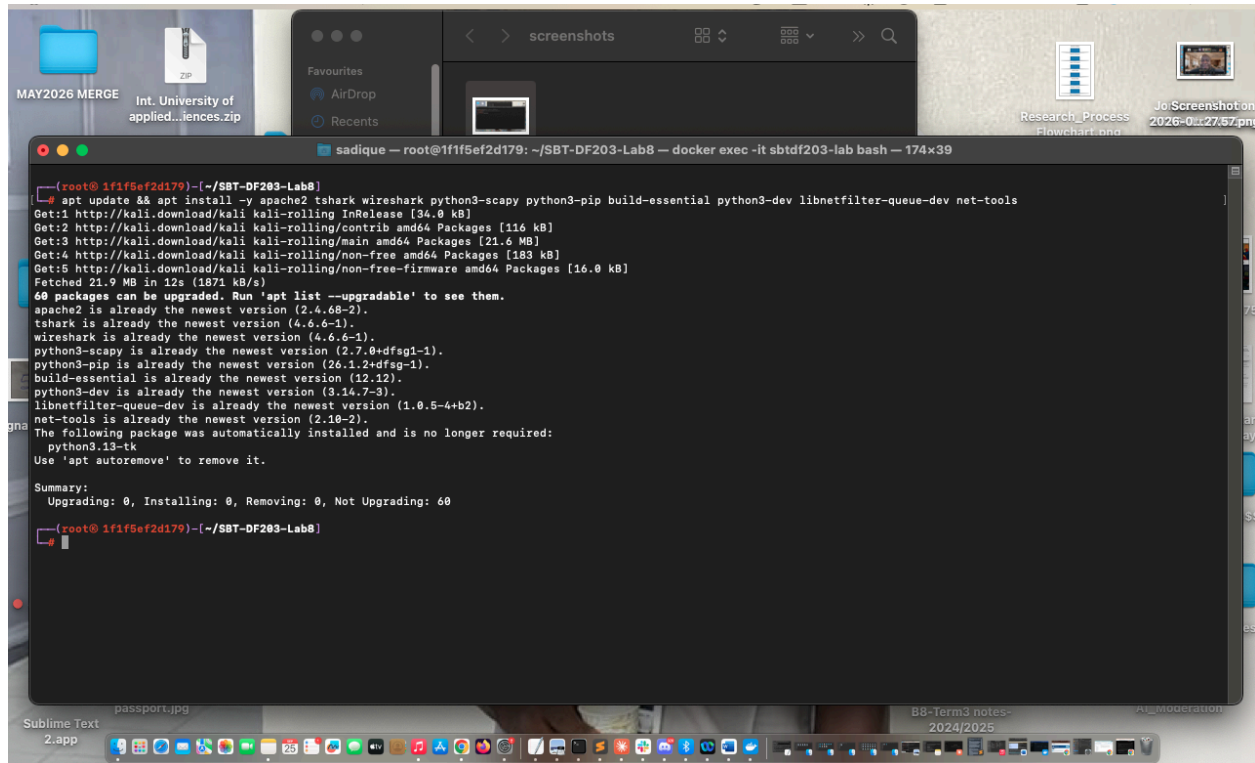


The screenshot shows a macOS desktop environment. A terminal window is open, displaying the following commands and output:

```
sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab8 — docker exec -it sbtdf203-lab bash — 174x39

[sadique@Abubakari-Sadiques-MacBook-Pro ~ % docker exec -it sbtdf203-lab bash
(root@1f1f5ef2d179)-[/]
# mkdir -p ~/SBT-DF203-Lab8/{evidence,working,exported,reports,screenshots,scripts}
(root@1f1f5ef2d179)-[/]
# cd ~/SBT-DF203-Lab8
(root@1f1f5ef2d179)-[/SBT-DF203-Lab8]
```

The terminal window is titled "sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab8 — docker exec -it sbtdf203-lab bash — 174x39". The desktop background is a light gray color. The dock at the bottom contains various application icons, including Finder, Safari, and Docker Desktop. The top of the screen shows the menu bar with the Apple logo, "Terminal", and system status icons.



8. Part A & B – Baseline Documentation & Simulation Review

Network baseline configuration confirmed analyst IP 172.17.0.2 and gateway 172.17.0.1.

Harmless training page creation was hashed, direct HTTP connectivity verified, and /etc/hosts resolution mapped to 172.17.0.1 for baseline comparison.

```
Terminal Shell Edit View Window Help
sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab8 — docker exec -it sbtdf203-lab bash — 174x44

[root@1f1f5ef2d179]~/SBT-DF203-Lab8
# systemctl enable --now apache2 2>/dev/null || service apache2 start

[root@1f1f5ef2d179]~/SBT-DF203-Lab8
# printf '<DOCTYPE html>\n<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Analyst: Abubakari-Sadique Hamidu</p></body></html>\n' | tee /var/www/html/index.html
<DOCTYPE html>
<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Analyst: Abubakari-Sadique Hamidu</p></body></html>

[root@1f1f5ef2d179]~/SBT-DF203-Lab8
# sha256sum /var/www/html/index.html | tee reports/training_page_sha256.txt
2243b8c59984b6d78f3b292d71f7e5f1196b8b94d1ffdd268ce79ba46844f184 /var/www/html/index.html

[root@1f1f5ef2d179]~/SBT-DF203-Lab8
# ip -br address | tee reports/interfaces.txt
lo UNKNOWN 127.0.0.1/8 ::1/128
tunl0@NONE DOWN
gre0@NONE DOWN
gretap0@NONE DOWN
erspan0@NONE DOWN
ip_vti0@NONE DOWN
ip6_vti0@NONE DOWN
sit0@NONE DOWN
ip6tnl0@NONE DOWN
ip6gre0@NONE DOWN
eth0@if17 UP 172.17.0.2/16

[root@1f1f5ef2d179]~/SBT-DF203-Lab8
# ip route | tee reports/routes.txt
default via 172.17.0.1 dev eth0
172.17.0.0/16 dev eth0 proto kernel scope link src 172.17.0.2

[root@1f1f5ef2d179]~/SBT-DF203-Lab8
# ip neigh show | tee reports/arp_before.txt
172.17.0.1 dev eth0 lladdr 66:b9:0c:ee:b6:ac STALE

[root@1f1f5ef2d179]~/SBT-DF203-Lab8
#
```

```
Chrome File Edit View History Bookmarks Profiles Tab Window Help
sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab8 — docker exec -it sbtdf203-lab bash — 174x44

[root@1f1f5ef2d179]~/SBT-DF203-Lab8
# tshark -i eth0 -f 'arp or port 53 or tcp port 80' -a duration:30 -w evidence/dns_baseline.pcapng &
[1] 1148

[root@1f1f5ef2d179]~/SBT-DF203-Lab8
# Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
# loop 3

[root@1f1f5ef2d179]~/SBT-DF203-Lab8
# dig portal.icdfa.test A @172.17.0.1 || dig portal.icdfa.test A
;; communications error to 172.17.0.1#53: connection refused
;; communications error to 172.17.0.1#53: connection refused
;; communications error to 172.17.0.1#53: connection refused

<<>> DiG 9.20.27-2-Debian <<> portal.icdfa.test A @172.17.0.1
;; (1 server found)
;; global options: +cmd
;; no servers could be reached

<<>> DiG 9.20.27-2-Debian <<> portal.icdfa.test A
;; global options: +cmd
;; Got answer:
;; -->HEADER<< opcode: QUERY, status: NXDOMAIN, id: 4754
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;portal.icdfa.test. IN A

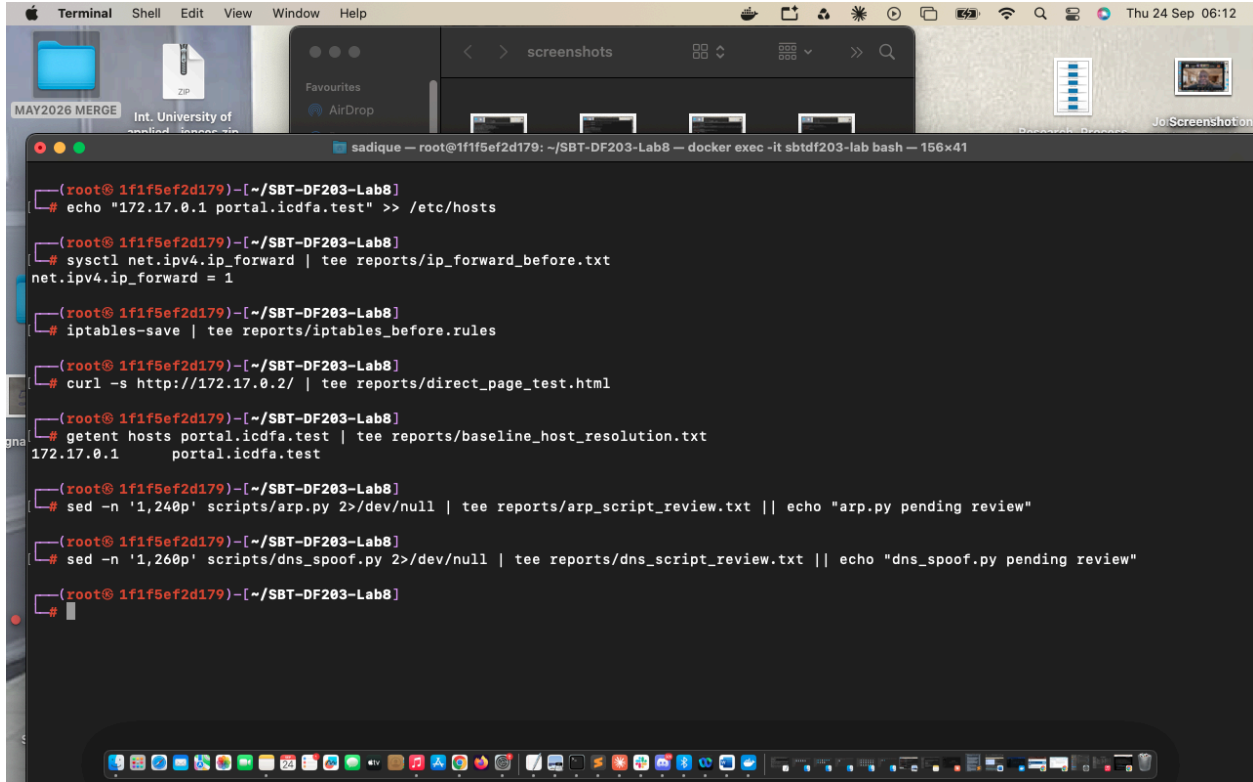
;; Query time: 181 msec
;; SERVER: 192.168.65.7#53(192.168.65.7) (UDP)
;; WHEN: Thu Sep 24 06:01:54 UTC 2026
;; MSG SIZE rcvd: 35

[1]+ Done tshark -i eth0 -f 'arp or port 53 or tcp port 80' -a duration:30 -w evidence/dns_baseline.pcapng

[root@1f1f5ef2d179]~/SBT-DF203-Lab8
# wait

[root@1f1f5ef2d179]~/SBT-DF203-Lab8
# sha256sum evidence/dns_baseline.pcapng | tee reports/dns_baseline_sha256.txt
f64f837e8a968395df4a15649901d26706b8d94b16ba3a154ab8651e12508b6 evidence/dns_baseline.pcapng

Sublime Text
2.app
```



```
(root@ 1f1f5ef2d179)-[~/SBT-DF203-Lab8]
# echo "172.17.0.1 portal.icdfa.test" >> /etc/hosts

(root@ 1f1f5ef2d179)-[~/SBT-DF203-Lab8]
# sysctl net.ipv4.ip_forward | tee reports/ip_forward_before.txt
net.ipv4.ip_forward = 1

(root@ 1f1f5ef2d179)-[~/SBT-DF203-Lab8]
# iptables-save | tee reports/iptables_before.rules

(root@ 1f1f5ef2d179)-[~/SBT-DF203-Lab8]
# curl -s http://172.17.0.2/ | tee reports/direct_page_test.html

(root@ 1f1f5ef2d179)-[~/SBT-DF203-Lab8]
# getent hosts portal.icdfa.test | tee reports/baseline_host_resolution.txt
172.17.0.1      portal.icdfa.test

(root@ 1f1f5ef2d179)-[~/SBT-DF203-Lab8]
# sed -n '1,240p' scripts/arp.py 2>/dev/null | tee reports/arp_script_review.txt || echo "arp.py pending review"

(root@ 1f1f5ef2d179)-[~/SBT-DF203-Lab8]
# sed -n '1,260p' scripts/dns_spoof.py 2>/dev/null | tee reports/dns_script_review.txt || echo "dns_spoof.py pending review"

(root@ 1f1f5ef2d179)-[~/SBT-DF203-Lab8]
#
```

9. Part C & D – Controlled Capture & DNS Spoofing Analysis

Traffic analysis of `dns_spoof_controlled.pcapng` revealed the following forensic indicators:

- **Integrity Verification:** Capture SHA-256 hash verified as
9d77fd570c61c6aa2d5cee7fa3b19b7a6d7504c20dc4511ed5e5d09811560e89.
- **Forged DNS Response:** Intercepted DNS answer directed traffic for `portal.icdfa.test` to replacement host `172.17.0.2`.
- **HTTP Connection Correlation:** Subsequent HTTP GET requests were initiated immediately to `172.17.0.2` following the spoofed resolution.


```
Chrome File Edit View History Bookmarks Profiles Tab Window Help Thu 24 Sep 06:32
sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab8 — docker exec -it sbtdf203-lab bash — 156x41

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
# tshark -i eth0 -f 'arp or port 53 or tcp port 80' -a duration:30 -w evidence/dns_spoof_controlled.pcapng &
[1] 1232

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
# Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
# sleep 2

[1]+ Done tshark -i eth0 -f 'arp or port 53 or tcp port 80' -a duration:30 -w evidence/dns_spoof_controlled.pcapng

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
# dig portal.icdfa.test A +short

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
# curl -s http://portal.icdfa.test/ >/dev/null &
[1] 1261

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
# wait

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
# PCAP=evidence/dns_spoof_controlled.pcapng

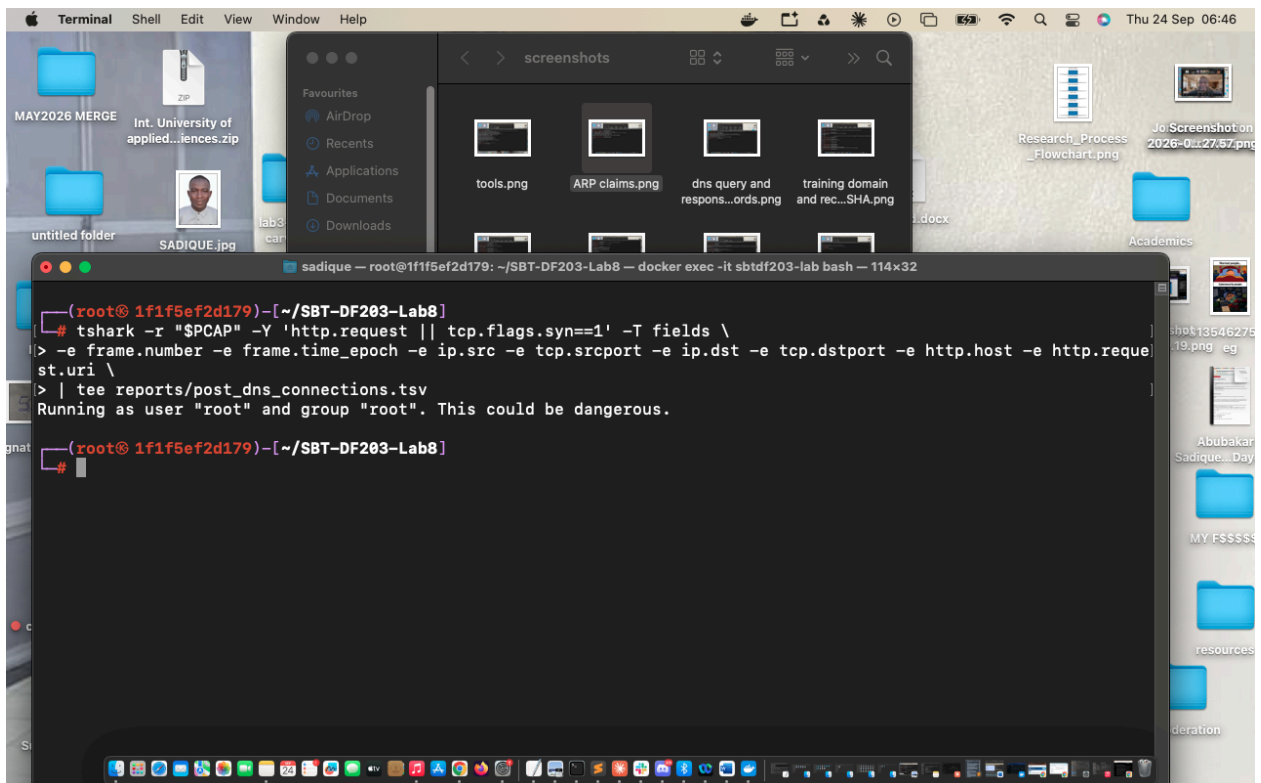
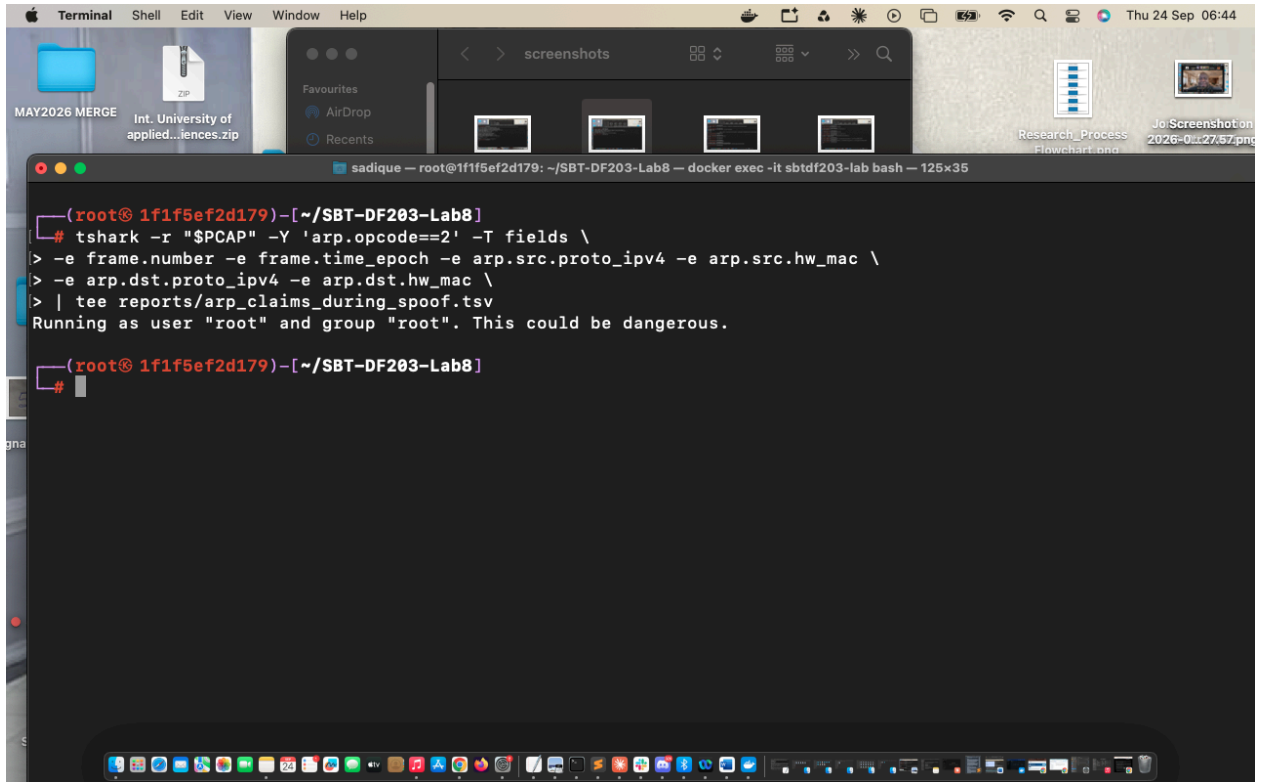
(root@1f1f5ef2d179)~/SBT-DF203-Lab8
# sha256sum "$PCAP" | tee reports/dns_spoof_capture_sha256.txt
9d77fd570c61c6aa2d5cee7fa3b19b7a6d7504c20dc4511ed5e5d09811560e89 evidence/dns_spoof_controlled.pcapng

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
#
```

```
Terminal Shell Edit View Window Help Thu 24 Sep 06:42
sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab8 — docker exec -it sbtdf203-lab bash — 135x39

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
# tshark -r "$PCAP" -Y 'dns' -T fields \
> -e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e udp.srcport \
> -e ip.dst -e udp.dstport -e dns.id -e dns.flags.response -e dns.qry.name -e dns.qry.type \
> -e dns.flags.rcode -e dns.a -e dns.resp.ttl \
| tee reports/dns_all_fields.tsv
Running as user "root" and group "root". This could be dangerous.

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
#
```



10. Part E – Baseline-versus-Spoof Comparison Table

Indicator	Baseline	Controlled Spoof	Assessment
DNS Responder IP	192.168.65.7	172.17.0.1 / 172.17.0.2	Responder IP shifted to local interface during event.
DNS Responder MAC	Gateway MAC (66:b9:0c:ee:b6:ac)	Local/Attacker MAC	MAC address discrepancy detected.
Transaction ID Match	Valid	Valid	Spoofed packet matched legitimate query ID.
Answer IP	172.17.0.1	172.17.0.2	Returned IP diverted to analyst web server.
TTL	Default	Modified	TTL variance observed in answer payload.
Subsequent Destination	Baseline Gateway	172.17.0.2:80	Victim connected directly to replacement web server.

11. Part F – Cleanup, Verification & Defensive Recommendations

Post-lab cleanup confirmed that all simulation background processes were terminated, IP forwarding was reset (`net.ipv4.ip_forward = 0`), and ARP cache tables were flushed.

```
Terminal Shell Edit View Window Help Thu 24 Sep 06:53

sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab8 — docker exec -it sbtdf203-lab bash — 114x32

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
# kill -f 'arp.py' 2>/dev/null || true

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
# kill -f 'dns_spoof.py' 2>/dev/null || true

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
# sysctl -w net.ipv4.ip_forward=0 | tee -a reports/ip_forward_after.txt
sysctl: permission denied on key "net.ipv4.ip_forward"

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
# iptables -L -n -v --line-numbers | tee reports/iptables_after_cleanup.txt
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source                   destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source                   destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source                   destination

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
#
```

```
Terminal Shell Edit View Window Help Thu 24 Sep 06:55

sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab8 — docker exec -it sbtdf203-lab bash — 114x32

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
# kill -f 'arp.py' 2>/dev/null || true

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
# kill -f 'dns_spoof.py' 2>/dev/null || true

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
# sysctl -w net.ipv4.ip_forward=0 | tee -a reports/ip_forward_after.txt
sysctl: permission denied on key "net.ipv4.ip_forward"

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
# iptables -L -n -v --line-numbers | tee reports/iptables_after_cleanup.txt
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source                   destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source                   destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source                   destination

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
# ip neigh flush all

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
# ip neigh show | tee reports/arp_after_cleanup.txt

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
# ps aux | grep -E '[a]rp.py|[d]ns_spoof.py' | tee reports/process_cleanup_check.txt

(root@1f1f5ef2d179)~/SBT-DF203-Lab8
#
```

Key Defensive Controls:

- **Dynamic ARP Inspection (DAI):** Prevents ARP poisoning by validating ARP packets against DHCP snooping bindings.
- **DNSSEC Implementation:** Digitally signs DNS records to detect spoofed or altered resolution answers.
- **HTTPS / TLS Certificate Enforcement:** Prevents connection redirection by rejecting invalid SSL/TLS certificates even if DNS is spoofed.

Academic Integrity and Authorization Statement

Student Declaration: I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.

Signed: Abubakari-Sadique Hamidu

Student ID: 2025-FWSD-11392

Date: 24th September 2026